ENERGY AND THE ENVIRONMENT

# Policy Responses to Climate Change

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- The human enhancement of global warming leading to climate change is seen as a worldwide problem.
- Policy responses have been led by international negotiation, but have been qualified or indecisive at the national level, and so far largely ineffective, despite strong international agreement on the matter.
- The principal focus is on reducing carbon dioxide emissions.
- Nuclear power is seldom acknowledged as the single most significant means of limiting the increase in greenhouse gas concentrations while enabling access to abundant electricity.

Emissions of greenhouse gases have a global impact, unlike some other forms of pollution. Whether they are emitted in Asia, Africa, Europe, or the Americas, they rapidly disperse evenly across the globe. This is one reason why efforts to address climate change have been through international collaboration and agreement.

The principal forum for international climate change action has been the United Nations, which has led to the Framework Convention on Climate Change (UNFCCC) and the Kyoto Protocol. In December 2015 the Paris agreement consolidated years of negotiations with agreement among 188 countries to limit carbon dioxide emissions. It came into force in 2016.

Privacy - Terms

Although climate change agreements emphasizing carbon emission reduction have been reached through international approaches, the policy measures to meet the obligations and objectives set by such agreements have been implemented at the national or regional level. Here they are supplemented by policy instruments such as efficiency standards and incentives to invest in infrastructure which does not give rise to carbon emissions. Pricing carbon emissions is seen as putting a price on a major external cost from energy production and transformation.

## The UN climate change negotiations, early phase

In 1988 the World Meteorological Organization (WMO) and the United Nations Environment Programme (UNEP) set up the Intergovernmental Panel on Climate Change (IPCC), an expert body that would assess scientific information on climate change. As a reaction to the concerns raised in the IPCC's First Assessment Report the UN General Assembly established the Intergovernmental Negotiating Committee for a Framework Convention on Climate Change. The UN Framework Convention on Climate Change (UNFCCC) was adopted in May 1992 and entered into force in 1994. The convention included the commitment to stabilize greenhouse gas emissions at 1990 levels by 2000.

The first Convention of the Parties to the UNFCCC (COP 1) was held in 1995. Negotiations at this and two subsequent COPs led to agreement on the Kyoto Protocol in 1997. The Kyoto Protocol set out specific commitments by individual developed countries to reduced emissions by an average of 5.2% below 1990 levels by the period 2008-2012. However, it would take three further meetings until the "Marrakesh Accords" were agreed, which provide sufficient detail on the procedures for pursuing objectives set out in the Kyoto Protocol.

The 1997 Kyoto Protocol involved several decisions:

- By 2012, developed countries would reduce their collective emissions by 5.2% from 1990 levels, each country being committed to a particular figure.
- The emissions covered by the Protocol are not only carbon dioxide, but also methane, nitrous oxide, hydrofluorocarbons, perfluorocarbons and sulfur hexafluoride.

- These commitments would be reckoned on a net basis, considering sinks as well as sources, and each country must credibly measure its contribution and meet its commitment.

- Countries may fulfil their commitments jointly (such as with regional agreements) and they may improve the efficiency of compliance through "flexibility mechanisms".

In order for the Kyoto Protocol to enter into force and become legally binding it had to be ratified by at least 55 countries and for those ratifying countries to include enough Annex I (developed) countries to represent at least 55% of the total emissions from those Annex 1 countries in 1990. In 2001 the US Government (which had earlier signed the Protocol) announced that it would not ratify the Protocol. As the USA emits more than a quarter of all greenhouse gas emissions from developed countries, this put the ratification of the Protocol in jeopardy. Australia also declared that it would not ratify, though it would pursue emission reductions as agreed.

Eventually, entry into force depended on the decision of Russia, another large greenhouse gas emitter. After some delay Russia notified the United Nations of its decision to ratify the Protocol in November 2004 and 90 days later, on February 19, 2005, the Protocol finally came into force. Australia subsequently ratified the Protocol in December 2007.

While countries that are party to the Protocol are expected to rely mainly on reducing their own emissions domestically, three "flexibility mechanisms" were identified to improve the economic efficiency of reductions and make it easier for parties to comply. The three mechanisms are emissions trading, Joint Implementation and the Clean Development Mechanism.

Emissions Trading: A market-based approach to achieving environmental objectives that allows those countries or entities reducing greenhouse gas emissions below what is required to use or trade the excess reductions to offset emissions at another source, inside or outside the country. In general, trading can occur at the domestic, regional (EU), international and intra-company levels. A precedent is the USA acid rain program, which successfully trades permits for sulfur dioxide.

Joint Implementation (JI): A project-based mechanism, whereby one developed country – with emissions caps – can work with another to reduce emissions or enhance sinks, and share the resulting emission reduction units accordingly.

The Clean Development Mechanism (CDM): A project-based mechanism where certified projects proposed by developed countries – or companies from those countries – can be used to reduce emissions in developing countries. The developed country – or company – earns certified emission reduction units, which may be used against the country's own reduction commitment. CDM is primarily focused on development aid and secondly on emission reduction.

## The Kyoto Protocol and nuclear energy

The role of nuclear energy in combating climate change received a lot of attention during the UNFCCC negotiations between COP 4 and COP 7 (1998-2002). This was due to the entrenched anti-nuclear position of some of the environment NGOs lobbying at the negotiations and the tendency for national delegations to be dominated by those from Environment Departments, with a historically more negative position towards nuclear energy than their overall national position.

Nuclear energy is discriminated against within the Marakesh Accords, specifically within the sections dealing with the Clean Development Mechanism and Joint Implementation, but currently the effect of this discrimination is largely symbolic.

The Marakesh Accords state:
"Recognizing that Parties included in Annex I are to refrain from using credits (from CDM or JI projects) generated from nuclear facilities to meet their commitments under Article 3, paragraph 1"
This text is convoluted, reflecting perhaps a compromise reached during the negotiations. It should be noted that CDM and JI projects involving nuclear facilities are not banned. Parties are free to put forward such projects, as they would do any other candidate project.

However, the text says that developed counties (Annex I) Parties should refrain from using any credits earned from those projects for meeting their commitments – which are the emissions targets agreed under the Kyoto Protocol. The meaning of "should refrain" is a matter of debate. Annex I Parties are also meant not to exceed their emissions targets. Should they refrain from using nuclear project credits unless it means they would miss their target?

Ultimately, this is a symbolic discussion, as the current low price of CDM and JI credits and the short time period over which credits would be awarded mean that the availability of such credits is

unlikely to be a significant factor in the decision on whether to invest in nuclear energy.

Concerns over the efficacy of CDM and JI in general have been expressed, many potential investors in projects being frustrated by the bureaucratic process involved in gaining approval for a project and the relatively small rewards for doing so. Moreover, there are concerns over the limited geographical distribution of CDM projects, with the majority of projects taking place in China, India and Brazil.

However, in the longer term the CDM and JI may become a more viable mechanism for encouraging low carbon projects and development. However, it is also possible that new mechanisms will be introduced in subsequent agreements and the role of the CDM and JI may diminish.

## Approaches to emissions trading and alternatives, European ETS

The question of emission permits of some kind as a basis for trading in them or trading them off has been approached in several different ways. They may be auctioned, or they can be allocated to firms on the basis of historical emissions (known as grandfathering). Within countries, emissions (e.g. carbon) taxes may be used rather than emissions trading, but still linked to the price of permits. An attractive feature of tradable permits is that any national scheme can be linked internationally. However, the emission caps need to be set by regulators, who have an impossible task in the light of normal uncertainties, as shown the first decade of the EU system. Also it tends to reward traders more than innovators.

The best-developed arrangement is the European Emissions Trading System (ETS), which was intended as the cornerstone of EU policy to counter climate change, but seems unlikely to deliver. The ETS is a cap-and-trade system, which means that a limited cap is set on the total amount of certain greenhouse gases, notably $CO_2$, that can be emitted by defined industry sectors. Within the cap, companies can receive or buy allowances that can be traded with one another. The cap is then linearly reduced over time so that total emissions decrease. From 2012 national caps became an EU-wide cap. The ETS is seen as providing the core of a wider scheme to limit carbon emissions worldwide. It aims to reduce Europe's emissions 40% below 1990 levels by 2030. It covers some 11,000 installations (power stations and industrial plants) in 28 EU

countries plus Norway, accounting for nearly half of the EU's carbon emissions. However, it does not count emissions from burning biomass, which have become large. All bioenergy is counted as zero-carbon inside the ETS, and there are policy incentives to replace coal with biomass as a means of increasing the share of renewables in the energy mix. With the trading of allowances, this results in more $CO_2$ emissions. In 2014, over 60% of renewable energy in Europe was bioenergy or waste, which helpfully complements intermittent and non-dispatchable wind and solar.

In 2011, carbon to the value of about €112 billion was traded on the ETS, but in 2012 this dropped to about €75 billion, its lowest level since 2008.

After a positive start in 2005, in May 2006 the price of emission allowances under the ETS for the first commitment period (2005-2007) plunged to less than half their previous value, causing intense discussion on the efficacy of the whole scheme and making it clear that the caps in some states were too low to promote investment in emission reduction. Most EU countries had issued so many allowances on the basis of padded applications that they did not reach their quotas in the first year of phase one (2005-07) of the ETS, which undercut the value of traded allowances. Allowances in mid-2006 were trading at €18/tonne $CO_2$, representing over 1.5 cents/kWh on coal-fired generation and providing a weak disincentive to using coal, especially in Germany where output constraints apply on nuclear power. For most of 2005 and until May 2006, permits were trading at over €25.

Overall in the EU, 1785 million tonnes of $CO_2$ were emitted in 2005 against quotas of 1829 Mt, though this did not necessarily represent any decrease from what emissions would have been in the absence of the EU ETS. The UK was 33 Mt (16%) over its quota, reflecting the low target set by its government, and a swing back to coal from gas. This meant that generators (particularly) in the UK needed to purchase allowances and pass the cost on to consumers. However, it was thought that many generators had already passed on much of the price of the carbon allowances allocated to them as an "opportunity cost".

In the second commitment/ trading period of the EU ETS (Phase II, 2008-2012) emissions allowance allocations were reduced 6.5% from those in the first commitment period. However the economic crisis radically altered the situation, and from 2009 the ETS had a growing surplus of allowances and international credits because the global economic crisis had depressed emissions more than anticipated, which significantly weakened the price signal.

Installations received trading credits from their national allocation plans (NAPs), administered by the governments of the 30 participating countries.

At one level it may be argued that the low price of emissions allowances in the first period could be considered as a success of the EU ETS, promoting the discovery of low-cost carbon avoidance measures. However, the credibility of the EU ETS as a part of broader climate change policies will depend on whether governments setting emissions allocations sufficiently tightly that they ensure the industries covered make a proportionate contribution towards meeting national targets, as part of the EU's overall target.

In January 2008 the European Commission (EC) proposed changes to the EU ETS in the third commitment and trading period 2013-20 which would strengthen and extend the scope of the trading scheme. NAPs would be replaced by centralized allocation by an EU authority and a single EU-wide cap on emissions which is to decrease by 1.74 % each year to 2020, when the cap would be 21% lower than the 2005 starting level. It was intended that the annual reduction would continue after 2020, with a review of the magnitude of the annual reduction to take place by 2025 at the latest. During the third commitment period a much larger share of emissions allowances would be auctioned instead of allocated free of charge. The scheme would broadened to include new industries (e.g. aluminium and ammonia producers) and new gases (nitrous oxide and perfluorocarbons). Due to the ETS having a growing surplus of allowances, the EC postponed the auctioning of some allowances as an immediate measure, and in November 2012 proposed other changes. These include increasing the EU's 2020 emission reduction target from 20% to 30%, making the annual 1.74% reduction steeper, retiring some Phase III allowances permanently, bringing more sectors into the ETS, and limiting access to international credits. These were considered in 2013.

In January 2014 the EC published its 2030 Framework for Climate and Energy Policies, including a legislative proposal for the ETS to establish a **market stability reserve (MSR)** to operate in the fourth commitment and trading period (Phase IV) running 2021 to 2030. The reserve would both address the surplus of emission allowances that has built up during Phase II and which keeps the carbon price very low, and also improve the system's robustness by automatically adjusting the supply of allowances to be auctioned. Together with postponing in auctions of 900 million allowances ("backloading"), the proposal was widely supported in addressing the problems facing the rather discredited ETS. In February 2014

backloading was approved with the withdrawal of 400 million allowances that year. In May 2014 the EC said it was open to introducing the market stability measures before 2021, since the oversupply remained large.

In February 2015 the European Parliament voted in favour of a market stability reserve (MSR) to operate from 2019, and for the 900 million surplus allowances to be added to it, along with 750 million unallocated allowances. In July 2015 the EC proposed putting about 250 million unallocated MSR allowances (from Phase II) plus 100 million unallocated from phase III into a phase IV new entrant reserve (NER), earmarked for new installations and significant capacity expansions. Any unused allowances due to plant closures or reduced production in phase IV will be added to the NER, making a likely total of about 395 million. It also proposed putting 50 million MSR allowances to top up an innovation fund of 400 million phase IV allowances sold by the European Investment Bank (EIB), for CCS, innovative and breakthrough technologies in energy-intensive industries. At €25/t this is worth €11.25 billion. These funding proposals are instead of keeping the MSR allowances off the market, or cancelling them.

For Phase IV, the EC also proposed increasing the rate of reduction after 2020 to 2.2% per year, in line with the 40% domestic greenhouse gas reduction target for the EU by 2030 (relative to 1990). It proposed auctioning 57% of a total 15.5 billion allowances available under the ETS cap for 2021-30, leaving 43%, about 6.3 billion, for free issue. The 57% is the same as in 2015. The original ETS directive envisaged the free allocation of allowances to diminish to zero in 2027. Analysts expect the ETS carbon price to reach at least €20 by 2020 and €30 by 2030. In February 2017 the European Parliament voted in favour of the EC's proposed revisions to the ETS from 2020, strengthening the market stability reserve (MSR), cancelling 800 million allowances in 2021 and supporting a 2.2% annual cap reduction. Subject to EU Council approval, this means the MSR will withdraw 24% of the surplus each year for four years instead of the EC's originally proposed 12% per year rate. European power industry group Eurelectric said the vote was "an important step towards strengthening the EU ETS and delivering a stronger carbon price in the short and medium term."

The shift from free allocation to auctioning of emissions allowances, as well as the tightening of emissions allowance caps, will benefit nuclear energy and other forms of low carbon generation. Although it is thought that the cost of emissions allowances has already been incorporated as an opportunity cost, the free allocation of

allowances means that fossil fuel power plant operators have not faced a true cost themselves for much of the emissions of their plant. In particular, when new electricity generation capacity is considered, fossil fuel generation will have to incorporate the cost of carbon allowances into the economic assessment of the plant. In addition, the tightening allowance cap is likely to lead to higher allowance prices, increasing the cost of greenhouse gas emissions.

However, the ETS has arguably exported rather than reduced $CO_2$ emissions. While the UK's carbon emissions fell some 15% over 1990 to 2005, when imports were taken into account, carbon emissions on a world basis attributable to UK went up more than 19%, according to Professor Dieter Helm. The general trend for western countries, which substitute imports from countries like China for domestic production, is obvious. The EC is addressing this carbon leakage issue with some free allowances, and in July 2015 considered creating four "at risk" categories, from very high to low, eligible for different levels of free allowances.

Elsewhere a simple carbon emission tax has been applied (contentiously in Australia, at a level far above EU ETS but still too low and with too many exemptions to drive change, before being abandoned). This is straightforward, but the level set is arbitrary and may be out of kilter with ETS systems. Also, like the ETS, it exempts certain industries on an arbitrary or politically-driven basis, creating major inequities. And like the ETS it penalizes exporters while exempting imports.

A carbon consumption tax has been proposed, which is applied like a normal consumption tax but on the basis of $CO_2$ implications in the production of goods and services. This would require certification of sources, not just at manufacturing level but also upstream to power generation, and at the border for imports (where assumptions re electricity inputs would need to be made). It would be equitable between domestic production and imports. Professor Dieter Helm in the UK is a proponent of this, but it is not implemented anywhere.

## Early UNFCCC negotiations

The protracted delays before the eventual entry into force of the Kyoto Protocol meant that the UNFCCC negotiations began to consider what emissions reduction regime should follow the first commitment period, which ended in 2012.

Discussions followed two parallel tracks, the first considering future emissions reduction commitments for Annex I Parties to the Kyoto Protocol, the second considering long-term global cooperative action to address climate change. This second track had a more sensitive task, as it included the USA, without emissions reduction targets for the first commitment period, and also developing countries, which did not have emissions reduction targets.

Decisions on appropriate action beyond 2012 generated a great deal of controversy. Emissions from developing countries were expected to rise as they increased their use of fossil fuels to meet their need for energy for development. Many developed countries wanted developing countries to limit the growth in their emissions. Developing countries pointed out that their per capita emissions are still much lower than those of developed countries and believe that developed countries should show their commitment to reducing emissions first, before expecting developing countries to take action.

The COP 15 meeting in Copenhagen in December 2009 was intended to result in a high-level political agreement, with a more detailed agreement to follow, however an agreement of all parties was not reached. The Copenhagen Accord, drawn up in the last days of the conference, included non-binding declarations of emissions reduction policies and commitments from individual governments, but no international agreement.

Short-term and long-term focus of UNFCCC negotiations

The ultimate objective of the Framework Convention on Climate Change is the stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system. This is a long-term objective that will require emissions paths to be reduced progressively over the 21st century and beyond.

However, the Kyoto Protocol focused attention on a relatively short-term emissions objective, namely the first commitment period between 2008 and 2012. The targets set for this period were first agreed in 1997, which gave governments 10-15 years to put in place policies to reach these targets, and only three years between the Protocol's entry into force in 2005 and the start of the first commitment period. This bias towards short-term targets did not provide any incentives to make the investments in long-term infrastructure changes, such as energy, transport and buildings, which are needed to bring sustainable reductions in greenhouse gas emissions.

UNFCCC conferences have grown from modest gatherings to the largest annual conferences held under UN auspices. The 2021 COP26 conference in Glasgow is expected to consolidate some of the positive intentions arising from COP21 in Paris.

A useful database of $CO_2$ emissions per capita is EDGAR (Emissions Database for Global Atmospheric Research).

UN climate change agreement December 2015, Paris

The major emphasis of the November-December 2015 COP21 meeting in Paris was on producing a global, binding agreement to cut carbon emissions. At the Paris meeting there was clear international agreement that reducing carbon dioxide emissions was a global priority built on a groundswell of public opinion in many countries, albeit with a range of different timelines involved. It was agreed to aim for a temperature increase well below 2 °C and with the aim of moving to 1.5 °C, which made it clear that governments would have to introduce additional mitigation actions to move more rapidly to low-carbon technologies, especially in electricity generation. The main and widely recognized implication (which fuelled some extravagant hype stigmatising coal) is that more use must be made of low- or zero-carbon energy sources, including nuclear power.

The International Energy Agency (IEA) described it as "nothing less than a historic milestone for the global energy sector" that would "speed up the transformation of the energy sector by accelerating investments in cleaner technologies and energy efficiency." With wide support, especially from likely beneficiaries, a clean energy innovation fund was to be set up under the UNFCCC's Green Climate Fund (GCF) to develop cleaner, more affordable and more reliable energy sources. Also the GCF should "aim to ensure efficient access to financial resources through simplified approval procedures" for developing countries. Whatever the advances in electricity storage associated with intermittent renewables, there is now more clearly an inexorable logic for low-cost continuous reliable supply from expanded nuclear power. The IEA had already made it plain that keeping the global temperature increase below 2°C would require a significant contribution from nuclear energy.

Agneta Rising, Director General of the World Nuclear Association said: "We welcome the commitments that governments have made, and the nuclear industry stands ready to help achieve the goals of the Paris agreement. This agreement should lead to a more positive outlook for nuclear investments, as nuclear is an important part of the response to climate change in countries across the

world. What governments need to do now is convert the global agreement they have reached in Paris into national policies, including a progressive decarbonization of the electricity generation sector. We have proposed that there should be 1000 GWe of nuclear new build by 2050 as part of a balanced low-carbon future energy mix. To achieve this, we need to see the introduction of energy markets with level playing fields which recognize the value of low carbon and reliable generation. We need to see the adoption of harmonized nuclear regulatory processes internationally. We also need to ensure that actions do not lead to clean nuclear power plants being closed prematurely and replaced with more polluting alternatives. Ongoing investment is also needed to help develop the next generation of nuclear technology, along with a clear and achievable pathway for deployment."

Ahead of COP21, 188 nations had submitted their individual climate action plans, including how much they were intending to cut emissions. There is a wide range of targets in these Intended Nationally Determined Contributions (NDCs), from ambitious cuts by 2030 to almost doubling emissions by 2030, according to individual national circumstances. Collectively the NDCs, if met, are projected to result in a global temperature rise above pre-industrial levels of 2.7°C, which is considered insufficient constraint. National targets are not binding, there are no defined sanctions for failing to meet them, and they need verification anyway as well as five-yearly reviews ratcheting up the good intentions.

First stocktaking talks were in 2018. Countries with targets for 2025 should have new targets for 2020, while those with 2030 targets are invited to update them. This process is to be repeated every five years, with a first post-2020 review in 2023. The agreement requires countries regularly to update climate commitments, with each pledge being more ambitious than the last. It also invites countries to write long-term, mid-century low carbon emission strategies by 2020. In contrast to the Kyoto Protocol, the agreement binds all countries equally to these processes (but not targets), though developed countries are expected to "continue taking the lead by undertaking economy-wide absolute emission reduction targets."

In the power sector, 70% of additional electricity generation to 2030 would be low-carbon. The full implementation of these pledges will require the energy sector to invest $13.5 trillion in energy efficiency and low-carbon technologies from 2015 to 2030, an annual average of $840 billion, according to the IEA. Excluded

from all this is the role of forest and peat fires in contributing to emissions.

The Paris Agreement was made available in April 2016 for signature and ratification by individual countries. Following EU ratification, early in October 2016 it had been ratified by countries responsible for 55% of global greenhouse gas emissions, and by over 55 of the 197 signatories. This implies consent to be bound by the terms of the agreement. It came into legal force, to the limited extent envisaged, on 4 November 2016. The USA and China – together representing 40% of global emissions – ratified the agreement together in early September. However, the new US administration in 2017 took the view that the agreement was essentially a treaty, which should have been ratified by the Senate. In June 2017 President Trump announced that the USA would withdraw from the Paris Agreement. His successor in 2020 reversed this. In any case, several US states and cities along with many business leaders said they would continue to pursue the Paris goals – California and New York in particular. There was to be no further US contribution to the UNFCCC's $100 billion Green Climate Fund – it had pledged $3 billion and paid $1 billion. By April 2021 the fund had $8.3 billion in total.

Subsequent COP meetings have done little to advance the Paris Agreement.

# COP28

World Nuclear Association, in partnership with Emirates Nuclear Energy Company (ENEC), the 'host nuclear company' of the COP28 meeting in Dubai, set up the Net Zero Nuclear initiative to encourage greater industry participation in the COP process, and to support countries that were seeking to establish their own declaration on nuclear energy.

In December 2023 at the COP28 meeting, governments from 22 countries signed a declaration calling for the tripling of global nuclear energy capacity from 2020 by 2050. The number of supporting countries grew to 25 before the end of the COP28 meeting.

Also at COP28, over 120 nuclear companies signed the Net Zero Nuclear Industry Pledge, which commits to a goal of at least tripling nuclear capacity by 2050.

Subsequently, the 'tripling goal' was supported at the March 2024 Nuclear Energy Summit organized by the Belgium government and the International Atomic Energy Agency, as well as at the G7 summit in Italy in June 2024.

Parties to the Paris Agreement had agreed to periodically take stock of the implementation of the agreement to assess progress towards achieving its goals. The first global stocktake, which was agreed at the COP28 meeting, included a paragraph that called for deep, rapid and sustained reductions in greenhouse gas emissions through measures including: “Accelerating zero- and low-emission technologies, including, inter alia, renewables, nuclear, abatement and removal technologies such as carbon capture and utilization and storage, particularly in hard-to-abate sectors, and low-carbon hydrogen production.”

This was the first time that nuclear energy had been mentioned in a positive context in a major COP agreement.

## COP29

At the COP29 meeting, developed countries agreed to take the lead in providing $300 billion a year in climate finance by 2035, with other countries encouraged to also contribute. Additionally, there was a vaguer call to raise $1.3 trillion per year from a wider range of sources, including the private sector. The $1.3 trillion figure was what developing countries had said would be required to address the impacts of climate change.

COP29 also saw agreement being reached on details for carbon markets, which had first been outlined in Article 6 of the Paris Agreement, ten years previously.

One area of the COP29 negotiations where agreement was not reached was on how to take forward the pledges made in the first global stocktake. A draft document that was in development was forwarded for further negotiation at COP30.

Outside of the formal COP29 negotiation process six countries were added to the Declaration to Triple Nuclear Energy, bringing the total number of supporting countries to 31.

COP29 also saw bilaterial agreements on nuclear energy being signed by the USA and Ukraine on SMR development, and the USA and UK on advanced nuclear reactors.

Also at the COP29 meeting was an announcement by the US government of a roadmap to add 200 GWe of new capacity by 2050, which would result in a domestic tripling of nuclear capacity. That 2050 target has since been raised to a quadrupling of nuclear capacity by the new administration, requiring around 300 GWe of new capacity, although the US government has also withdrawn from the UNFCCC process.

## Europe

In many respects Europe has been a leader in promoting action on climate change, as set out in some detail above.

In March 2007 the European Council endorsed the European Commission's Strategic Energy Review and agreed on a unilateral cut of 20% in EU greenhouse gas emissions by 2020, relative to 1990 levels. The previous commitment was 8% reduction by 2012. This required strengthening and extending carbon trading arrangements as well as deploying low- or zero-carbon technology. The European Council also endorsed the objective of making a 30% reduction in greenhouse gas emissions by 2020 and said that it would commit to this 30% target if other developed countries committed to (unspecified) comparable reductions in emissions and the more advanced developing countries (e.g. India, Brazil, China) "contributed adequately according to their responsibilities and respective capabilities". French President Chirac described the outcome as "one of the great moments of European history."

The European Council also set a target of meeting 20% of EU energy needs from renewables by 2020, leaving individual countries to decide their own policies in such a way as to allow nuclear power as part of their energy mix to be taken into consideration in allocating individual country targets for renewables. The Council noted "the European Commission's assessment of the contribution of nuclear energy in meeting the growing concerns about safety of energy supply and $CO_2$ emission reductions" and it acknowledged the role of nuclear energy "as a low $CO_2$-emitting energy source." In the event the 2008 policy set was “20-20-20” – 20% reduction in CO2 emissions, 20% of electricity from renewables and 20% improvement in energy efficiency by 2020.

The European Commission’s 2030 Policy Framework for Climate and Energy in January 2014 moved away from major reliance on renewables to achieve emission reduction targets and allows scope

for nuclear power to play a larger role. It is focused on $CO_2$ emission reduction, not the means of achieving that, and allows more consideration for cost-effectiveness.

The centerpiece is a binding 40% reduction in domestic greenhouse gas emissions by 2030 (compared with a 1990 baseline) which will require strong commitments from EU member states. Current policies and measures if followed through should deliver 32% reduction by then, so 40% "is achievable" and widely supported. It implies a 43% cut from 2005 for $CO_2$ in sectors covered by the EU emissions trading scheme (ETS). There are to be no post-2020 national renewables targets, and individual states are free to use whatever technology they wish to achieve emission reductions in the longer term, though a 27% "headline target at European level for renewable energy" is included. The framework also proposes reform of the ETS to make it the principal driver of climate policy (see Emission Trading section above), and it drops a binding energy efficiency target and a directive for use of biofuels in transport.

Impetus for the profound change in emphasis from the 2008 policy framework appears to have come from EU member states which are winding back renewables programs due to escalating costs. The International Energy Agency has pointed out the huge difference in energy prices between USA and EU, with gas prices three times as high and electricity twice as high in the EU. The EU is evidently concerned about loss of international competitiveness and the increasingly chaotic retreat from subsidy schemes related to its 2020 renewables target. More generally, it acknowledges that "the rapid development of renewable energy sources now poses new challenges for the energy system".

The key change from 2020 goals is "providing flexibility for Member States to define a low-carbon transition appropriate to their specific circumstances, preferred energy mix and needs in terms of energy security, and allowing them to keep costs to a minimum." An early test of this will be approval for UK plans to set long-term electricity prices to enable investment in nuclear plants.

The WNA said that the "flexible" approach outlined allows nuclear power to play an expanded role in decarbonising electricity supply. The ambitious target "is a bare minimum if the EU wishes to achieve its objective of an 80% reduction by 2050, and do its part in averting a 2°C rise in global temperatures. Unfortunately the target of 27% for renewable energy continues to undermine the possibility for cost efficiency in meeting the carbon target. It also again demonstrates an unjustified preference in EU policy for renewable energy over other carbon reduction pathways – such as

nuclear energy – regardless of cost, maturity and the preferences of individual Member States."

However, only weeks later the EU parliament in a non-binding resolution voted by 341 to 263 to claw back some of the previous provisions by changing the EC draft policy to call for binding national targets of 30% of power from renewables (not 27% overall) and reinstating the energy efficiency goal to 40% improvement by 2030, along with the EC 40% greenhouse gas reduction. Member states can however go with the EC draft policy rather than this.

## Asia Pacific Partnership and Clean Energy Ministerial

The Asia-Pacific Partnership on Clean Development and Climate, known informally as APP, was a non-treaty partnership established by Australia, India, Japan, China, South Korea and the United States and launched in 2006. The Partnership involved countries that account for about half of the world's population and more than half of the world's economy, energy use, and greenhouse gas emissions. In October 2007, Canada joined APP.

The objectives of APP included measures to work towards cleaner, more efficient energy technologies, goods, and services in key sectors. It envisaged collaboration on many aspects of energy supply, including nuclear power.

In April 2011 APP wound up, and was superseded by the Clean Energy Ministerial (CEM), a high-level global forum that had been established in 2010 following the Copenhagen COP, with a number of collaborative initiatives. The Clean Energy Ministerial includes representatives from 26 governments plus the EU, representing 70% of global GDP and 80% of global greenhouse gas emissions. A significant CEM initiative is the 21st Century Power Partnership. Another is Nuclear Innovation: Clean Energy Future.

## Contraction and Convergence

The concept of Contraction and Convergence (C&C) is a long-term framework towards the ultimate object of climate change policy in terms of 'safe' emissions levels. The concept has gained some interest amongst politicians and climate change experts and is

seen as potentially superseding the arbitrary short-term target setting of the Kyoto Protocol process.

Under a Contraction and Convergence regime an international agreement would define to what level atmospheric greenhouse gas concentrations could rise before becoming unacceptable. Once this is defined, an estimate would be made of how much reduction in global greenhouse gas emissions is required to meet the target, taking into account the effect of sinks, and how quickly the target should be reached. This represents the 'contraction' element, and in itself it does not differ substantially from the aims of the UNFCCC to stabilize "greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system."

The key differentiating factor of Contraction and Convergence is the proposal that ultimately the 'right' to emit carbon dioxide is a human right which should be shared equally. Therefore, emissions targets should ultimately be allocated to countries on the basis of their populations. Emissions rights would be on a per capita basis and therefore require convergence from the present very unequal per capita levels to a universal per capita level.

During the convergence period, which should not be protracted, emission permits would be progressively adjusted from status quo to these new levels. Permits could be traded, and this would cause a major economic transfer from countries that have used fossil fuels to create wealth to those still struggling to alleviate poverty. After convergence, each country would receive the same allocation of carbon dioxide emission rights per head of population and further trading in permits is envisaged.

Carbon emissions stocktake

Global emissions of carbon dioxide increased from 22.7 billion tonnes in 1990 to 36.4 billion in 2019, a rise of 60%. China's emissions increased from 2.42 to 10.17 billion tonnes over this period, India's from 0.58 to 2.62 billion tonnes. The USA increased from 5.13 to 5.28 billion tonnes and the EU declined from 3.87 to 2.92 billion tonnes.

Apart from policies to use low- or zero-carbon sources for electricity generation, some countries have favoured the use of natural gas to replace coal, on the basis that emissions from actually burning the fuel are around half of those from coal. However, methane leakage from the drilling and pipeline delivery of natural gas can offset any $CO_2$ benefits that natural gas may bring over coal during combustion and use. A 3% leakage of natural gas will push the

global warming effect of natural gas used for electricity to the same level as that of coal per kWh.

# References

United Nations Framework Convention on Climate Change webpage on the Green Climate Fund

# Related information

Climate Change - The Science
Electric Vehicles
Renewable Energy and Electricity
Hydrogen Production and Uses
Carbon Dioxide Emissions From Electricity